

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Gloves need to be worn when using hand warmers☐ Boiling water is used to recharge battery powered hand warmers☐ Some chemical reactions give out heat energy☒ All hand warmers are reusable☐			1	1		1
		(ii)		award (1) for each of following cheapest accept cheap / only costs £1 least temperature drop (over time) / keeps warmer longer neutral answer – it lasts longer			2	2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		award (2) for all points plotted correctly – tolerance $\pm\frac{1}{2}$ square award (1) for any 6 points plotted correctly award (1) for smooth curve through all points (from origin)		3		3	3	
		(ii)		2 hours ☐ 3 hours ☒ 4 hours ☐ 5 hours ☐		1		1	1	1
		(iii)		Iron reacts with oxygen forming iron oxide until all the oxygen is used up ☐ Heat formed expands the iron ☐ Iron oxide loses oxygen, forming iron ☐ Iron reacts with oxygen forming iron oxide until all the iron is used up ☒			1	1		1
				Question 7 total	0	4	4	8	4	5